

PRODUCT REQUIREMENTS

Product Requirements Document

RERA Scrapper — MahaRERA Due-Diligence & Company Charter Pipeline

Integrow Asset Management

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Field	Value
Document	Product Requirements Document
Product	RERA Scraper
Version	1.0
Date	11 August 2026
Status	Baseline — documents the product as built, plus forward roadmap
Owner	Integrow Asset Management
Companion documents	SAD.md (architecture), CLAUDE.md (flow), rules.md (content rules), guardrails.md (guards)

1. Executive summary

RERA Scraper turns a single MahaRERA registration number into a defensible, source-cited due-diligence pack in one command.

Real-estate underwriting in Maharashtra requires assembling a picture of three things — the **counterparty** (the promoter company), its **promoters** (the individuals behind it), and the **collateral** (the project and its land). That picture is scattered across a regulator's portal, three corporate-registry mirrors, two credit-rating agencies, an insolvency board, a GST portal, a Marathi-language land-records system, and the open web. Assembling it by hand takes an analyst days, produces inconsistent output, and — most importantly — leaves no audit trail of what was checked versus what was found.

This product automates that assembly and, critically, **governs what the resulting document is allowed to say**. It produces a paired **Company Charter** — an Internal variant that keeps every process failure and full bibliographic sourcing, and an External variant that is client-shareable with numbered citations — plus a RERA project summary PDF, a Developer Score, a Documentation Confidence Score, and a numbered, actionable gap list.

The product's distinguishing characteristic is not that it scrapes. It is that **it refuses to ship a document it cannot show to be compliant**, and that it never invents a fact to fill a hole.

ONE COMMAND	THE PACK
-----	-----
python main.py P51800012345	+---> Company_Charter_..._Internal.pdf
[--gstin X --pan Y]	Company_Charter_..._External.pdf
	+---> Company_Charter_...facts.json
	<REG_NO>_summary.pdf
1 human CAPTCHA solve	documents/ + complaint_orders/
(+1 per GST lookup)	usage_summary.json (cost ledger)
	+---> a numbered, actionable gap list

2. Problem statement

2.1 The user's problem

An underwriter evaluating a Maharashtra real-estate opportunity must answer, defensibly:

Question	Where the answer lives today
Is this developer real, or a single-project vehicle?	MahaRERA promoter search + past-experience records, manually cross-read
What is their track record — delivered area, on-time rate, complaints?	Every one of their other registered projects, one page at a time
Who actually controls the company?	ZaubaCorp, Tofler, InstaFinancials — which frequently disagree
Are they insolvent, or in proceedings?	IBBI public search

Question	Where the answer lives today
Are they rated? By whom?	ICRA and Infomerics, separately
Are they filing GST on time?	GST portal, one CAPTCHA per lookup
Is the land what they say it is?	Maha Bhulekh Property Card — Marathi-only, CAPTCHA per call
Is the project litigated?	MahaRERA complaints, appeals, orders and judgments
What is the micro-market actually doing?	The open web, of highly uneven quality
What did we NOT manage to establish?	Nowhere. This is the question nobody records.

That last row is the real problem. A manual diligence pack tells you what was found. It rarely tells you, in a form an investment committee can act on, **what was looked for and not found** — and it almost never distinguishes "we checked and it is clear" from "we looked and could not establish this." Those are completely different risks.

2.2 The failure modes this product exists to prevent

FAILURE MODE -----	PRODUCT RESPONSE -----
"No litigation found against the promoter, the project or the underlying land in any source reviewed; MahaRERA's records are empty[18]; the Title Report's 30-year search returned nothing[1]"	Section B: A CLEAN CHECK PRODUCES NO SENTENCE. Three citations spent establishing there is nothing to report is three citations wasted. The line is DELETED. The scope of what was checked stays in the record and does not reach the page.
Two registries name different directors; the tool silently picks one	BOTH are surfaced as a roster conflict and become a gap that renders in BOTH document variants. Neither is picked.
A source is cited that does not actually support the claim	"A missing marker is a lesser failure than a wrong one: leave a clause uncited rather than borrow an adjacent source."
A number is quietly guessed to fill a table cell	Anything unconfirmable goes in `gaps`, verbatim. Two known data contradictions are DELIBERATELY left unreconciled.
A model hallucinates and the document is blocked (or shipped) on that basis	Every violation the reviewer reports is verified against the document text first. An invented quote cannot block a good document.
An API outage silently deletes a real finding	run_finding_research: a researcher that raises costs ONLY its own finding, and no finding is ever deleted whatever the researcher returns.
The document gets cleaner while the record gets thinner every run	Scrub/restore contract: the page drops, the record keeps. .facts.json always holds the complete text.

3. Background and context

3.1 Regulatory context

The Real Estate (Regulation and Development) Act, 2016 requires every qualifying project in India to register with a state authority and publish a defined public record. **MahaRERA** is Maharashtra's authority. The public record includes project details, promoters and associated partners, professionals of record, uploaded documents, complaints, appeals and SRO details.

That record is *public*, but it is not *accessible*: two of nine data categories are open, and the other seven sit behind a short-lived guest session that the portal mints only after a human solves a text CAPTCHA on the project detail page.

Two further public systems matter and are similarly gated:

- **GST portal** (services.gst.gov.in) — a taxpayer's return-filing history, one CAPTCHA per lookup.
- **Maha Bhulekh** (bhulekh.mahabhumi.gov.in) — the Property Card (मालमत्ता पत्रक) land record, Marathi-only, with a CAPTCHA that regenerates on every partial postback and therefore admits **no reusable session at all**.

3.2 Organisational context

Integrow Asset Management underwrites Indian real-estate exposures. The Company Charter is the artefact its investment process consumes. Two audiences exist for the same underlying analysis:

- **Internal** — the underwriting team, who need the process failures, the file names, the unverified items and the full bibliography.
- **External** — the counterparty or co-investor, who need a clean, numbered-citation document with no internal tooling detail in it.

Producing two documents by hand from one analysis is where inconsistency creeps in. **Producing both from one facts dictionary, forking only at render time, is the product's core structural bet.**

3.3 Product history

Date	Milestone
—	Initial scraper: resolve, auth, 9 categories, summary PDF
—	Agentic deep research added (deep_research.py), with per-claim verification and bounded gap retry
—	First Company Charter (Pranami Bliss) produced by hand-written Node/docx-js scripts
—	company_charter.py replaces those scripts: template-filled in place, dual variant
—	Promoter portfolio, MCA-mirror chain, credit rating, IBBI, judgments search
—	GST intake and compliance scoring; CTS land-record path
2026-07-31	GST intake confirmed live end-to-end: PAN → GSTIN → 76 scoreable filing periods spanning 2022-08 to 2026-06
2026-08-10	Charter reshape shipped (all 7 tasks of CHARTER_RESHAPE_SPEC.md): flags-vs-gaps split, clean-check deletion, per-clause External citations, facts-schema corrections, per-finding deep research. charter_document.py 's builder retired. Rules migrated from CLAUDE.md into rules.md
2026-08-11	Guardrails documentation and the blocking compliance review hardened; this PRD baselined

4. Goals and non-goals

4.1 Product goals

#	Goal	Measurable expression
PG1	Compress diligence time from days to one command	One operator, one command, one CAPTCHA solve produces the full pack
PG2	Every claim traceable	100 % of External factual claims carry a [N] marker resolving to a checkable Sources entry
PG3	Never present a guess as a fact	Zero fabricated values; unconfirmable items appear as numbered gaps
PG4	Distinguish "clear" from "unknown"	Clean checks produce no sentence; gaps keep their full explanation
PG5	A partial failure never costs the run	Every optional stage degrades to a warning; the run completes
PG6	A non-compliant document does not ship	The PDF is blocked unless the document can be <i>shown</i> to comply

#	Goal	Measurable expression
PG7	Two audiences, one analysis	Internal and External render from the same facts dict, forking only at render time
PG8	Costs are visible and arguable	Per-label token and USD cost in every run summary and <code>usage_summary.json</code>
PG9	Re-runs are cheap and diffable	Documents reused, research reused within 24 h, prior runs archived not clobbered
PG10	The record outlives the page	<code>.facts.json</code> retains everything the rendered document drops

4.2 Non-goals

#	Non-goal	Rationale
NG1	Making the investment decision	The product produces evidence and scores; a human underwrites
NG2	Solving CAPTCHAs	Hard ethical and ToS boundary, stated in three modules. A human solves every one
NG3	Becoming a general-purpose Indian RERA tool	Maharashtra only. Other states have different portals, schemas and land systems
NG4	Real-time monitoring	Batch, on-demand, human-triggered. Change detection is run-to-run, not continuous
NG5	Replacing legal title diligence	The land-record check is corroborative. A Title Report is a lawyer's product
NG6	A hosted multi-tenant service	Single-operator, local filesystem, Windows-bound PDF conversion
NG7	Reconciling genuinely contradictory sources	Two known contradictions are deliberately preserved. Picking a side would present a guess as a fact
NG8	Automating the CTS→village resolution	Marathi office and village labels do not map 1:1 to RERA's English text. Picking the wrong village would misattribute an entire legal land record

5. Success metrics

5.1 Primary metrics

Metric	Definition	Target
Time to pack	Command issued → Internal PDF on disk	< 30 min excluding human CAPTCHA wait
Human touches per run	CAPTCHA solves + interactive picks	1 (no GST) / 1 + 1 + N (with GST, N = GSTINs)
Citation coverage, External	Factual claims carrying a supporting <code>[N]</code> marker	100 % (was ~30 % pre-reshape, ~83 % mid-reshape)
Gate pass rate	Runs reaching PDF without a compliance block	> 95 %, with every block genuinely justified
Category completeness	Categories fetched / 9	9/9 with a session; 2/9 is a valid degraded outcome
Cost per pack	<code>usage_summary.json</code> total <code>cost_usd</code>	Tracked per run; no hard cap, but per-label attribution is mandatory

5.2 Quality metrics

Metric	Definition	Target
Clean-check leakage	Absence statements surviving into a rendered document	0
Cited absences	[N] markers attached to a nothing	0 (hard-gated)
Orphan sources	Sources listed but never cited	0
Mis-citations	A marker that does not support its clause	0 tolerated; a missing marker is preferred to a wrong one
Duplicate reporting	The same fact or absence appearing in two places	0
Findings lost to tooling failure	Findings deleted by a failed research call	0 — architecturally guaranteed
Guard-symbol drift	Symbols named in guardrails.md that no longer exist	0 (test-enforced)

5.3 Leading indicators of trouble

Indicator	What it means
--verify reports a confirmed endpoint failing	MahaRERA changed something load-bearing
Rising verification_error gap count	Anthropic API instability, or an expiring key
roster_conflicts appearing frequently	Registry mirrors drifting apart — a <i>feature</i> signal, not a bug
truncated: true on portfolios	PROMOTER_PROJECT_LIMIT (25) is biting; the score may understate
Compliance blocks clustering on one rule	That rule's deterministic pass is under-implemented

6. Users and personas

+=====+		
	PERSONA 1 -- THE ANALYST (primary user, ~90% of runs)	
+=====+		
	Goal Produce a diligence pack for a named project, today.	
	Context Windows laptop, Word installed, at the keyboard.	
	Inputs A registration number, sometimes only a project name, sometimes a PAN from the promoter's own document set.	
	Behaviour Runs `python main.py <REG>`, solves one CAPTCHA, walks away, comes back to a PDF pair.	
	Needs Speed. A clear run summary. To know what FAILED and how to retry just that piece.	
	Pain if A partial failure kills the whole run and wastes the unserved CAPTCHA solve.	
+=====+		
	PERSONA 2 -- THE UNDERWRITER / IC MEMBER (document consumer)	
+=====+		
	Goal Decide. Quickly, and defensibly.	
	Context Reads the Internal PDF; may forward the External one.	
	Needs Overview & Flags on page one. A Developer Score with its workings shown. A numbered gap list they can hand back as questions for the developer.	
	Pain if Two pages of "nothing was found" prose before the first unserved real finding. A score they cannot interrogate.	
+=====+		
	PERSONA 3 -- THE COUNTERPARTY / CO-INVESTOR (External doc reader)	
+=====+		
	Goal Understand what was assessed and on what basis.	
	Needs Numbered citations they can actually check. Descriptive source labels, not internal filenames.	
	Must NEVER A file path, module name, JSON key, raw exception string, see an internal tooling failure, or a bare CIN in prose.	
+=====+		
	PERSONA 4 -- THE ENGINEER (maintainer)	
+=====+		
	Goal Change one thing without breaking three.	
	Needs CLAUDE.md for flow, rules.md for content, guardrails.md for guards. Symbols not line numbers. A free, no-API render loop for verification.	
	Pain if A guard that looks redundant and is not. Stale docs that unserved describe a builder that was deleted.	
+=====+		
	PERSONA 5 -- THE ORCHESTRATOR (LLM agent, for charter_report.py)	
+=====+		
	Goal Run the "Counterparty + Collateral" document, which needs per-director web research an agent must spawn.	
	Needs A clean prepare()/finish() seam that returns prompts and accepts parsed replies.	
+=====+		

7. User journeys

7.1 Journey A — the standard run (Analyst, happy path)

ANALYST	SYSTEM	ARTEFACTS
-----	-----	-----
python main.py P51800012345 ----->	resolve via public search (no login, no CAPTCHA) <-- "[OK] Resolved to internal project ID: 23600"	
	no cached session <-- opens a VISIBLE browser	
solves the CAPTCHA ----->	token minted, cached 90 min --> .guest_token_cache <-- "[OK] Using session"	
[walks away]	archive previous run fetch 9 categories (8 in parallel) download documents (8 workers, reuse prior) download complaint orders build promoter portfolio agentic deep research (reuses confirmed sources if < 24 h old) generate Company Charter 5-way parallel checks editorial passes render Internal, External *** COMPLIANCE REVIEW *** convert both to PDF build summary PDF write usage log	----> output/_history/... ----> raw/*.json ----> documents/*.pdf ----> complaint_orders/ ----> promoter/portfolio.json ----> research/deep_research.json ----> company_charters/* ----> <REG>_summary.pdf ----> usage_summary.json
	<-- Run summary: auth source, per-category counts, documents downloaded, promoter_profile, market_research, gst_filing_intake, company_charter, claude_api_usage by label	
[returns]		====> THE PACK

7.2 Journey B — the run with GST (one extra decision, several extra solves)

```

ANALYST                                SYSTEM
-----                                -
Finds the promoter's PAN in
MahaRERA's own document set
(RERA always includes a PAN
card and never a GSTIN --
which is why --pan is the
flag you usually have)
  |
  v
python main.py P518... --pan AANCP0234D
  |
  v
                                ... stages 1-6 as above ...
                                |
                                v
                                STAGE 7 -- GST INTAKE
                                format-validate the PAN FIRST
                                (a typo must never cost a human
                                a CAPTCHA solve)
                                |
solve CAPTCHA #2 <----- search_gstins_by_pan
                                -> every GSTIN under the PAN,
                                across every state
                                |
solve CAPTCHA #3..N <----- fetch_gstin_filing_table per GSTIN
                                ONE solve then walks ALL financial
                                years in the same session
                                |
                                v
                                parse GSTR-1 / GSTR-3B periods only
                                (CMP08, GSTR9/9C, GSTR1A deliberately NOT
                                parsed: no statutory due-date rule is
                                implemented, so scoring them would be a
                                guess, not a computation)
                                |
                                v
                                primary GSTIN = the one with the MOST periods
                                -> gst_filing_input.json
                                |
                                v
                                STAGE 9: run_gst_compliance_check reads it
                                -> statutory due dates -> delays
                                -> GST Compliance sub-metric (7.5% weight)
                                -> flags if late_pct or recent delays breach
                                    _FLAG_THRESHOLDS

IF ANYTHING FAILS: one unscored sub-metric. NEVER the run.

```

7.3 Journey C — the land-record path (human-in-the-loop, four steps)

WHY THIS IS MANUAL:

Maha Bhulekh's office and village names are Marathi-only and do NOT map 1:1 to RERA's English district/taluka/village text. Confirmed live: one real project's OFFICE was "...Andheri" while its own VILLAGE was "Aambivali" -- one office's jurisdiction covers several villages. Picking the wrong village would misattribute an entire legal land record to the wrong place. So the system NEVER guesses a label.

```
STEP 0  python company_charter.py <REG>    (at least once, to have facts)
        | run_cts_land_lookup auto-fetches office candidates once
        | per reg_no, and re-emits the reminder gap on EVERY run
        | until cts_lookup_input.json exists -- so this cannot go
        | quiet and get forgotten after the first mention
        v
STEP 1  python cts_resolve.py offices <REG>    [headless, no CAPTCHA]
        -> cts_office_candidates.json
        -> ANALYST PICKS the office
        v
STEP 2  python cts_resolve.py villages <REG> "<office>" [headless]
        -> cts_village_candidates.json
        -> ANALYST PICKS the village
        v
STEP 3  python cts_resolve.py candidates <REG> "<office>" "<village>" <cts>
        -> cts_number_candidates.json [headless]
        -> ANALYST PICKS the exact CTS sub-division
        v
STEP 4  python cts_resolve.py finalize <REG> "<office>" "<village>" <cts> <mobile>
        -> cts_lookup_input.json {district, office, village,
                                cts_number, mobile}
        v
STEP 5  python main.py <REG>    (or company_charter.py <REG>)
        -> run_cts_land_lookup finds the input file
        -> opens a VISIBLE browser
        -> ANALYST SOLVES THE CAPTCHA
        -> Property Card scraped (+ OCR mar+eng, + screenshot)
        -> facts["cts_land_record_check"]
        -> *** CTS cross-checked against RERA's own
            survey_cts_plot_numbers; a genuine mismatch becomes
            facts["cts_mismatch_note"] and is promoted to an
            IMMINENT flag ***
```

7.4 Journey D — pre-RERA intake and later attachment

A CIN or a CTS number often reaches the desk BEFORE a RERA number exists.

```
TODAY                                     LATER
-----                                     -----

python promoter_intake.py <CIN> "<name>"
-> MCA profile + IBBI + group companies
    + credit rating (needs the name)
-> output/_pending/<CIN>/promoter_profile.json
    *** written with the EXACT facts.json keys ***
    |
python cts_intake.py <district> <office>      |
                    <village> <cts> <mobile> |
-> output/_pending/<slug>/land_record.json   |
    key: cts_land_record_check              |
    |                                       |
    +-----+-----+
    |
    v
python attach_rera.py <case_id> <REG>
    COPIES (never moves) into output/<REG>/
    as *_carryover.json
    "an explicit action, never auto-matched"
    |
    v
python main.py <REG>
    reuses the promoter profile INSTEAD of re-running
        live CIN checks -> the 5-way fan-out skips four
        of its five branches
    reuses the land record INSTEAD of opening a new
        CAPTCHA browser
    cross-checks CTS -> IMMINENT flag on mismatch
```

7.5 Journey E — recovery

SOMETHING FAILED	DO THIS	COST
-----	-----	----
deep research failed	python deep_research.py <REG>	API only
charter failed	python company_charter.py <REG>	API only
only the PDF is wrong	python finalize_report.py <REG>	*** ZERO network, no browser, no CAPTCHA ***
search cannot find a project that still exists	python main.py <REG> --project-id N (the direct API can still return full data for a project the search box cannot find -- a site-side index gap)	
endpoint drift suspected	python main.py <REG> --verify	
session expired	python token_cache.py minutes-left	
rendering change to check need to diff two runs	the free no-API _fill_template loop already done: output/ history/<REG>/<ts>/	

8. Product scope — the feature map

RERA SCRAPER -- FEATURE MAP	
F1 ACQUISITION +-----+ F1.1 Project resolution (reg-no or free text) F1.2 Guest session management (90-min cache) F1.3 9-category scrape F1.4 Document download + reuse F1.5 Complaint-order download F1.6 Run archiving + diffing F1.7 Endpoint verification +-----+	F2 CORPORATE INTELLIGENCE +-----+ F2.1 MCA profile, 3-way cross-checked chain F2.2 Director roster merge + CONFLICT SURFACING F2.3 Group / affiliated cos F2.4 Credit rating (ICRA + Infomercials, side by side) F2.5 IBBI insolvency check F2.6 CIN-only pre-RERA intake +-----+
F3 TRACK RECORD +-----+ F3.1 Promoter portfolio (all registered projects) F3.2 Cross-project complaint / appeal totals F3.3 Delivered area + on-time rate F3.4 5 km micro-market influence (geocoded) F3.5 Lapsed / revoked flagging +-----+	F4 COLLATERAL +-----+ F4.1 Land identification F4.2 CTS -> Property Card (human-in-the-loop) F4.3 CTS cross-check -> IMMINENT flag on mismatch F4.4 FSI metrics + mortgage / charge (a LIVE RIGHT IS A FINDING) F4.5 Document library + OCR +-----+
F5 COMPLIANCE +-----+ F5.1 GST intake (PAN -> all GSTINs -> filing tables) F5.2 Statutory due dates (monthly + QRMP, state- specific) F5.3 Delay measurement + scoring F5.4 MahaRERA judgments search F5.5 Appeal cross-referencing F5.6 Review authenticity triage +-----+	F6 MARKET +-----+ F6.1 Agentic macro-market research F6.2 Micro-market / locality F6.3 Promoter external profile F6.4 Per-claim verification F6.5 Bounded gap retry with DIFFERENT strategies F6.6 Comparables + distances F6.7 Per-finding deep research F6.8 Source-trust registry +-----+
F7 SCORING +-----+ F7.1 Developer Score 3 buckets, 9 sub-metrics NEVER renormalises F7.2 Imminent-flag hard cap F7.3 Documentation Confidence 8 criteria, RENORMALISES F7.4 Flag classification imminent/structural/ monitor F7.5 KPI cards +-----+	F8 DOCUMENTS +-----+ F8.1 Company Charter Internal F8.2 Company Charter External F8.3 facts.json (full record) F8.4 RERA summary PDF F8.5 Counterparty+Collateral report (standalone) F8.6 Executive Briefing (8-page companion) F8.7 PDF conversion +-----+
F9 GOVERNANCE +-----+ F9.1 rules.md as live config F9.2 Preflight gate F9.3 External quality gate (11 checks) F9.4 Blocking compliance review + quote verification F9.5 Deterministic fallbacks F9.6 Scrub / restore contract F9.7 Never-fatal wrappers +-----+	F10 OPERATIONS +-----+ F10.1 CLI entry point F10.2 Streamlit UI (3 tabs) F10.3 Per-label cost ledger F10.4 Run summary F10.5 Offline PDF rebuild F10.6 Component retry commands F10.7 Token cache CLI +-----+

| F9.8 Guard-symbol test |
+-----+

8.1 Feature priority

Priority	Features	Rationale
P0 — the product does not exist without these	F1.1–F1.4, F8.1–F8.3, F8.7, F9.1–F9.4, F9.7	Acquisition, the Charter pair, and the gates that make it defensible
P1 — core value	F2., F3., F6.1–F6.5, F7. *, F9.5–F9.6, F10.1, F10.3–F10.4	The intelligence and the scores
P2 — high value, opt-in	F4.2–F4.3, F5.1–F5.3, F6.7–F6.8, F1.6, F10.5–F10.7	Each costs a human solve or is a refinement
P3 — adjacent deliverables	F8.4–F8.6, F10.2, F5.6	Separate documents and the UI

9. Functional requirements

Requirements are numbered **FR-*<area>-<n>***. **MUST / SHOULD / MAY** carry RFC-2119 force.

9.1 Acquisition (FR-ACQ)

ID	Requirement	Priority
FR-ACQ-01	The system MUST accept either a MahaRERA registration number matching <code>^P\d{11}\$</code> or a free-text project name	P0
FR-ACQ-02	On multiple name matches the system MUST present the candidates and let the operator pick; with no interactive terminal it MUST exit code 2 with instructions rather than guess	P0
FR-ACQ-03	Project-id resolution MUST NOT require a login or a CAPTCHA, and MUST NOT click the "View Details" link (which fires a JS confirm and lands on the gated page)	P0
FR-ACQ-04	The system MUST accept <code>--project-id N</code> to bypass search entirely, for projects the portal's own search index cannot find but whose API record still exists	P1
FR-ACQ-05	The system MUST obtain a guest session by opening a visible browser for a human to solve the CAPTCHA, and MUST NOT attempt to read or solve the CAPTCHA image	P0
FR-ACQ-06	The system MUST cache a valid session on disk and reuse it for 90 minutes , against a real token lifetime of roughly 100 minutes	P0
FR-ACQ-07	The system MUST accept <code>--token</code> (manually captured) and <code>--no-auto-auth</code> (skip the browser), and MUST complete the run in all four auth states: <code>explicit, cached, fresh_browser, none</code>	P0
FR-ACQ-08	The system MUST fetch all nine categories, <code>projects</code> first and alone (it carries the <code>userProfileId</code> that <code>past_experiences</code> requires), the remaining eight in parallel	P0
FR-ACQ-09	The system MUST normalise MahaRERA's response envelopes and MUST treat a 200 response containing only bookkeeping keys as empty, not as data	P0
FR-ACQ-10	On 401/403 for gated categories despite holding a session, the system MUST invalidate the cache, re-acquire a session, and re-fetch only the failed subset — exactly once	P0
FR-ACQ-11	The system MUST write one raw JSON file per category, including a <code>{"_error":...}</code> sentinel on failure, so <code>raw/</code> always has nine files	P0
FR-ACQ-12	The system MUST download project documents by POST to the DMS service with the bearer token, falling back to direct-URL scanning for shapes that do not match	P0
FR-ACQ-13	The system MUST NOT trust the DMS <code>Content-Type</code> header, and MUST verify that a purported JSON body actually parses before treating it as JSON	P0

ID	Requirement	Priority
FR-ACQ-14	The system MUST de-duplicate filenames by appending (2), (3) before the extension, because two genuinely different documents can share a filename	P0
FR-ACQ-15	The system MUST reuse documents from the previous run, matched by (document_id, source_filename) for DMS downloads and by original_url for direct URLs, and record status: "reused"	P1
FR-ACQ-16	The system MUST download complaint-order PDFs, and this MUST NOT be fatal	P1
FR-ACQ-17	The system MUST archive the previous run by moving it to output/_history/<reg>/<timestamp>/, resolving same-second collisions with a numeric suffix	P1
FR-ACQ-18	The system MUST read prior research <i>before</i> archiving and prior manifests <i>after</i> archiving	P1
FR-ACQ-19	The system MUST provide --verify mode that probes every endpoint and prints its configured trust level (confirmed/observed) alongside the empirical outcome	P2

9.2 Corporate intelligence (FR-CORP)

ID	Requirement	Priority
FR-CORP-01	The system MUST extract a CIN or LLPIN from the RERA record and label it correctly	P1
FR-CORP-02	The system MUST query three independent MCA mirrors (ZaubaCorp, Tofler, InstaFinancials) for the company profile	P1
FR-CORP-03	When director rosters agree , the system MUST merge them silently	P1
FR-CORP-04	When director rosters disagree , the system MUST surface both as roster_conflicts, promote them into facts["gaps"], and render them in both document variants. It MUST NOT silently pick one	P0
FR-CORP-05	The system MUST detect a paywalled mirror response rather than treating it as data	P1
FR-CORP-06	The system MUST query both ICRA and Infomerics for credit ratings and present them side by side	P1
FR-CORP-07	The system MUST check IBBI for insolvency proceedings and classify the hit	P1
FR-CORP-08	The system MUST source group/affiliated companies from ZaubaCorp only, and MUST corroborate them against InstaFinancials directorship counts rather than treating them as authoritative	P1
FR-CORP-09	The system MUST run these five checks in parallel (credit rating, IBBI, company profile, group companies, judgments search)	P1
FR-CORP-10	Each of these checks MUST be individually never-fatal	P0
FR-CORP-11	The system MUST support a CIN-only pre-RERA intake writing output/_pending/<CIN>/promoter_profile.json with the exact `facts.json` keys, so a later merge is a plain dict update	P2
FR-CORP-12	When a promoter carryover is attached, the system MUST skip the four live corporate checks it replaces	P2

9.3 Track record (FR-TRK)

ID	Requirement	Priority
FR-TRK-01	The system MUST discover every project registered under the promoter's name via MahaRERA's Promoters search tab	P1
FR-TRK-02	Fan-out MUST be capped at PROMOTER_PROJECT_LIMIT = 25, and truncation MUST NEVER be silent — a truncated field and an extra limitation entry are required	P1
FR-TRK-03	The system MUST aggregate complaints, appeals, delivered area and on-time completion rate across the portfolio	P1
FR-TRK-04	On-time classification MUST compare actual against original proposed completion date, and MUST count an unparseable pair into neither bucket	P1

ID	Requirement	Priority
FR-TRK-05	The system MUST geocode project locations via OSM Nominatim at no more than ~1 request/second, with a per-run cache	P1
FR-TRK-06	A project whose geocode fails MUST be dropped from the 5 km sum and MUST NOT be treated as "0 km away"	P0
FR-TRK-07	The subject project MUST be excluded from area totals by the entry's own registration number or exact name — never by the fetch registration number, which destroyed real data for single-project SPVs	P1
FR-TRK-08	The system MUST flag lapsed, revoked or cancelled projects	P1
FR-TRK-09	The portfolio MUST always emit its standing list of seven limitations	P1
FR-TRK-10	Portfolio build failure MUST be never-fatal	P0

9.4 Collateral and land (FR-LAND)

ID	Requirement	Priority
FR-LAND-01	The system MUST extract land identification (survey/CTS, village, taluka/district, pincode, gross/affected/net area), each with its source	P0
FR-LAND-02	The system MUST support a Maha Bhulekh Property Card lookup, opt-in and never auto-run	P2
FR-LAND-03	The system MUST NOT fuzzy-match Marathi office or village labels against RERA's English text; the operator MUST pick from the site's own option list	P0
FR-LAND-04	The system MUST provide a four-step human-in-the-loop resolution chain (offices → villages → candidates → finalize), of which only the final fetch requires a CAPTCHA	P2
FR-LAND-05	The system MUST re-emit the land-record reminder gap on every run until cts_lookup_input.json exists, so it cannot be forgotten after the first mention	P2
FR-LAND-06	The system MUST cross-check a retrieved CTS number against RERA's own survey_cts_plot_numbers , and MUST raise an IMMINENT flag on a genuine mismatch	P1
FR-LAND-07	A composite key of district + village + CTS number MUST be used for pending land records, because a bare CTS number is not globally unique	P2
FR-LAND-08	The system MUST merge mortgage_area and mortgage_lender into a single "Mortgage / charge on the land" field, preserving the finding that development agreements permit a mortgage even where none has been taken — a live right is a finding	P1
FR-LAND-09	The system MUST extract text from downloaded documents, falling back to OCR, and MUST degrade to an [OCR unavailable] marker rather than failing when Tesseract is absent	P1

9.5 Compliance and litigation (FR-COMP)

ID	Requirement	Priority
FR-COMP-01	GST intake MUST be opt-in via mutually exclusive --gstin / --pan	P2
FR-COMP-02	The system MUST format-validate the identifier before opening a browser, because a typo must never cost a human a CAPTCHA solve	P2
FR-COMP-03	The system MUST enumerate every GSTIN registered under the PAN, across every state	P2
FR-COMP-04	One CAPTCHA solve per GSTIN MUST cover every financial year for that GSTIN	P2
FR-COMP-05	The system MUST parse only GSTR-1 and GSTR-3B into scoreable records. CMP08, GSTR9/9C and GSTR1A MUST NOT be scored, because no statutory due-date rule is implemented for them and scoring them would be a guess	P2
FR-COMP-06	The primary GSTIN MUST be the one with the most filing periods; the others' raw data MUST be retained, not discarded	P2

ID	Requirement	Priority
FR-COMP-07	The system MUST NOT write an empty <code>records: []</code> file, because the file's existence signals "GST data was supplied"	P2
FR-COMP-08	Due dates MUST be computed per period (not per GSTIN), detecting monthly vs QRMP frequency from the period span, because a taxpayer can switch schemes at any quarter boundary	P2
FR-COMP-09	QRMP GSTR-3B due day MUST be derived from the GSTIN's own state code (22nd for Category X, 24th for Category Y)	P2
FR-COMP-10	A period whose frequency cannot be resolved MUST be counted as <code>unresolvable_frequency</code> and excluded from every other count, never assigned a guessed due date	P2
FR-COMP-11	A period not yet due MUST be excluded from all counts — that is not a gap, it is simply not due yet	P2
FR-COMP-12	GST failure MUST cost exactly one unscored sub-metric, never the run	P0
FR-COMP-13	The system MUST search MahaRERA's orders/judgments and cross-reference the results against the project's appeals, saving the judgment PDFs	P1
FR-COMP-14	The system SHOULD triage the authenticity of any operator-supplied <code>reviews.json</code> (rating polarisation, bursts, near-duplicates, one-hit-wonder reviewers, claim cross-referencing)	P3

9.6 Market research (FR-RES)

ID	Requirement	Priority
FR-RES-01	The system MUST produce three research blocks: <code>macro_market</code> , <code>micro_market</code> , <code>promoter_external</code>	P1
FR-RES-02	Every claim MUST carry an inline source marker resolving to a <code>sources</code> entry with a URL	P1
FR-RES-03	Every cited source MUST be independently re-verified by a separate pass	P1
FR-RES-04	A source whose verification could not run (missing key, network, rate limit) MUST be kept , with an honest gap appended stating it was not re-verified this pass	P1
FR-RES-05	A source that is unsupported or stale MUST be dropped into gaps with the reason	P1
FR-RES-06	Unresolved gaps MUST get bounded retries (max 2), each with a different strategy , and retry results MUST themselves be re-verified before merging	P1
FR-RES-07	A gap that survives retry MUST be reported as such, and MUST NOT be fabricated away to shorten the list	P0
FR-RES-08	Research MUST be reusable within a 24-hour window: confirmed sources carried forward untouched, only open gaps re-attempted	P1
FR-RES-09	Every finding that survives MUST earn a dedicated follow-up research pass establishing what it is, who is involved, when it arose, whether it is still live, and what it means for this project — capped at 8 calls	P1
FR-RES-10	A failed per-finding research call MUST keep the original finding text. No finding may ever be deleted by a research failure	P0
FR-RES-11	The system MUST maintain a cross-run source-trust registry, tallying open-web domains and auto-promoting at the 5th distinct project, with promotion notes appearing in the Internal document only	P2
FR-RES-12	Deep research failure MUST be never-fatal	P0

9.7 Scoring (FR-SCORE)

ID	Requirement	Priority
FR-SCORE-01	The system MUST compute a Developer Score across three buckets — Operational Strength (50), Financial Strength (20), Governance Strength (30) — and nine sub-metrics	P1

ID	Requirement	Priority
FR-SCORE-02	An unscored sub-metric's weight MUST NOT be redistributed or removed from the denominator. A promoter with less publicly-verifiable data must structurally score lower	P0
FR-SCORE-03	If any imminent flag exists, a grade of AAA or AA MUST be capped to A. It MUST NOT soften an already-lower grade	P1
FR-SCORE-04	The system MUST compute a Documentation Confidence Score over eight criteria, renormalising when a criterion is N/A — the opposite policy to the Developer Score, and deliberate	P1
FR-SCORE-05	Both score tables MUST render fully intact however many N/A rows they contain; they are scoring methodology, not fact tables, and are exempt from empty-row removal	P1
FR-SCORE-06	The External variant MUST NOT render the Weight column in either score table (hard-gated)	P1
FR-SCORE-07	GST scoring MUST use breakpoints shared with flag classification, so a flagged project cannot silently score AAA	P1
FR-SCORE-08	Two distinct unscored GST outcomes MUST be distinguished: "no input supplied" (a pending-input gap) versus "input supplied but no scoreable period"	P2
FR-SCORE-09	The system MUST classify flags as imminent, structural or monitor, assigning each gap a stable 1-based number	P1
FR-SCORE-10	facts["developer_score"] MUST be treated as an output of a prior run, never a valid input. Every consumer MUST recompute	P1
FR-SCORE-11	The system MUST render four KPI cards: Completion Slippage, Units Sold, Litigation Load, Land Title — each coloured by its own real severity	P1

9.8 Documents (FR-DOC)

ID	Requirement	Priority
FR-DOC-01	The system MUST produce a Company Charter in exactly two variants, Internal and External, from one facts dictionary, forking only at render time	P0
FR-DOC-02	Internal MUST render first, on the real facts dict, and MUST compute the scores that get persisted. External MUST render from a deep copy	P0
FR-DOC-03	The document MUST have exactly ten locked top-level sections in a fixed sequence. Content may be reshaped inside the flow; sections MUST NOT be reordered or renamed	P0
FR-DOC-04	The template .docx MUST be filled in place , preserving every style, colour and table already baked into it	P0
FR-DOC-05	The final deliverable MUST be PDF. The .docx remains on disk as the conversion input, and as the only output if conversion fails	P0
FR-DOC-06	Any script invoking _fill_template directly MUST call the PDF conversion itself; only run_company_charter does it automatically	P0
FR-DOC-07	Charter files MUST only ever be saved to output/company_charters/	P0
FR-DOC-08	The system MUST persist a .facts.json containing the complete, unscrubbed record	P0
FR-DOC-09	The system MUST produce a RERA project summary PDF, which is a different document from the Charter	P1
FR-DOC-10	The summary PDF MUST carry a data-reliability note listing every category using a non- confirmed endpoint	P1
FR-DOC-11	Body text MUST be justified; table cells MUST be centred only for short values, with narrative prose left-aligned	P1
FR-DOC-12	Table rows MUST NOT split across a page break, and a continued table MUST repeat its header row	P1
FR-DOC-13	Long URLs MUST NOT appear in narrow table columns; a cell carries a short label or an [N] marker, and the full URL lives once in Sources	P1
FR-DOC-14	The Internal document MUST carry a version log with elapsed end-to-end run time, total API cost and call count	P2
FR-DOC-15	The system SHOULD support a standalone "Counterparty + Collateral" report with its own independent quality gate	P3
FR-DOC-16	The system MAY produce an 8-page Executive Briefing that narrates only what the Charter already established, performing no new interpretation	P3

9.9 Operations (FR-OPS)

ID	Requirement	Priority
FR-OPS-01	The system MUST print a run summary covering auth source, per-category counts, documents, promoter profile, market research, GST intake, Charter status and API usage	P0
FR-OPS-02	The system MUST record token and USD cost per label , so each stage's cost is separately visible and arguable	P1
FR-OPS-03	Usage tracking MUST be reset at the start of each run, because the Streamlit process is long-lived	P1
FR-OPS-04	The system MUST support rebuilding the summary PDF from disk with zero network calls, no browser and no CAPTCHA	P2
FR-OPS-05	Every optional stage MUST be individually retryable from the command line	P1
FR-OPS-06	The system MUST offer a Streamlit UI wrapping the same functions, with no duplicated logic	P3
FR-OPS-07	The token cache MUST expose a CLI with shell-friendly exit codes (0 success, 1 stale/absent, 2 bad usage)	P2
FR-OPS-08	Cost figures for an unknown model MUST read 0.0 rather than a silently wrong number	P1

10. Content and editorial requirements

These are product requirements, not style preferences. They are the reason the documents are usable, and they are enforced in code. The authoritative statement is `rules.md`; this section states them as requirements with their enforcement points.

10.1 The governing principle

```

+=====+
|
|  REPORT WHAT WAS FOUND.  NEVER PAD WHAT WAS NOT.
|
|  "We checked and it is clear"    -> DELETED
|  "We looked and could not      -> KEPT IN FULL, as a
|    establish this"              numbered gap
|
|  These are different risks. The document must not blur them.
|
+=====+
```

10.2 Editorial requirements (FR-ED)

ID	Requirement	Enforcement
FR-ED-01	A clean check produces no sentence. No sentence, no citation marker, no list of what was searched, no named sources. Delete the line	Prompt + <code>_scrub_clean_checks</code> + <code>_is_cited_absence</code> gate
FR-ED-02	Never attach a citation marker to an absence. Three carve-outs only: an entire empty section, an absence that is the stated basis of a score, and a gap	Hard gate check #4
FR-ED-03	An empty section keeps its heading and one bare line — "Nothing found." No sources, no scope, no explanation, no marker	<code>_remove_empty_section_headings</code> preserves the heading
FR-ED-04	Findings first. Never open on a negative when the passage goes on to give real content. Applies to table cells exactly as to bullets	Prompt + review
FR-ED-05	Absence as evidence survives only where it is the stated basis of a number the reader would otherwise take on trust	<code>_is_carved_out</code> patterns

ID	Requirement	Enforcement
FR-ED-06	A gap is not an absence. Gaps & Sources keeps every item with its full explanation, including permanent ones. Never compressed under the clean-check rule	<code>_is_cited_absence</code> returns False for Gap N.
FR-ED-07	Deep research on every finding. Anything surviving as a finding earns a dedicated follow-up pass, not a one-line mention	<code>run_finding_research</code>
FR-ED-08	A live right is a finding. A contractual permission, option, charge or entitlement counts as found even if never exercised, because it remains exercisable	<code>_merged_mortgage_value</code>
FR-ED-09	Ruled-out items survive, in the right section. Something discovered and dismissed is a finding, not an absence. Age alone never rules an item out	<code>_relocate_merge_orders</code>
FR-ED-10	Say it once. A fact, and an absence, appears in exactly one place. A Summary line is a verdict plus a pointer, never a restatement	<code>_point_rera_landowner_at_identity_table</code> + review
FR-ED-11	Flags summarise, Gaps explain. A flag is a one-sentence headline ending in (Gap N); the full explanation lives once, under Gaps & Sources, numbered to match	<code>_flag_headline</code> + <code>_classify_flags</code> gap numbering
FR-ED-12	Internal keeps process failures, External does not. Never put a file path, module name, function or parameter name, JSON key, or raw exception string into either document; least of all External	<code>_sanitize_process_text</code> + <code>_externalize_prose</code>
FR-ED-13	Deep-dive vs nothing-found. A section with real findings gets expanded properly; a genuinely empty one collapses to the heading plus "Nothing found." Never padded, never supported with sources or search scope	Prompt + review
FR-ED-14	No jargon. Every initialism (CIRP, NCLT, IBC, DSRA, IGR, KMP, ROC, MCA, IOD, IOA, CHS, NCD, SRO, SPOC, IBBI, FSI, DCPR, MHADA, MRTTP, CC/OC...) MUST be expanded or plain-language once on first use per section, keeping the term itself. A plain-language pass, not a dumbing-down pass — precision on legal and registry facts must not be lost	<code>_expand_jargon_first_use</code>
FR-ED-15	Gloss raw status strings. A verbatim registry value (e.g. IBBI's "ASSIGNMENT NOT APPROVED YET") is quoted as-is and translated in the same sentence. Never present an opaque database string as a finding	Prompt + review
FR-ED-16	Bullets stay the default format. Consolidate fragmented bullets into one well-formed bullet. Sentence casing, proper entity capitalisation, complete sentences	<code>_consolidate_bullet_clauses</code> , <code>_fix_bullet_capitalization</code>
FR-ED-17	CIN/DIN placement. Never inline in prose, anywhere, except the three identity tables (Corporate/Promoter Identity, Current Directors & KMP, Company Registration Profile)	<code>_strip_inline_cin_din</code>
FR-ED-18	Empty table/row removal. Drop a whole table if every row is empty; drop a row only if every column is empty. Except the two score tables	<code>_remove_gap_rows</code> , <code>_remove_fully_empty_rows</code>

10.3 External-only requirements (FR-EXT)

ID	Requirement	Enforcement
FR-EXT-01	Numbered citations. Every inline factual claim resolves to a [N] marker: one marker per distinct claim, not one glued to the end of a multi-sentence passage	<code>_register_citation</code> , <code>_insert_marker_at_clause_end</code>
FR-EXT-02	Marker placement. A marker sits at the end of the clause it supports, not the end of the paragraph. [1] [2] [4] parked after the final full stop is the failure this rule exists to prevent. A marker never lands mid-word	<code>_insert_marker_at_clause_end</code>

ID	Requirement	Enforcement
FR-EXT-03	Scope is total. Executive Summary bullets, flag headlines, table cells, appendix prose and score-table notes are all in scope. A section shipping with no markers at all is a defect, not a stylistic choice	Advisory judge + review
FR-EXT-04	The marker must actually support the claim. Citing the MahaRERA complaints record for a sentence beginning "independent web research found..." is a mis-citation even though both are real sources. A missing marker is a lesser failure than a wrong one	<code>_clause_topic_citation</code> returns None rather than borrowing an adjacent source
FR-EXT-05	Descriptive source labels. Issuer, what the document is, and its date. Never a raw internal filename, never a bare category label a reader cannot check	<code>_external_source_label</code> , <code>_generic_one_label</code>
FR-EXT-06	No em dash, no hyphen-pair dash anywhere in an External paragraph	Hard gate checks #1 and #2
FR-EXT-07	Only gaps that also earned an Imminent or Structural flag appear in External, colour-coded by severity and numbered to match	<code>_external_gaps</code>
FR-EXT-08	The External Sources list is built only from tokens actually cited , in first-use order, so an uncited source can never appear	<code>_citation_registry</code> order
FR-EXT-09	Numbered Sources entries MUST retain balanced parentheses and their bullet numbering	Hard gate checks #7 and #8
FR-EXT-10	The Document Library table and the Weight columns MUST NOT appear in External	Hard gate checks #9 and #10; <code>t[8]</code> dropped

10.4 The consequence rule (a constraint on the rules themselves)

```

Sections B and C of rules.md are injected VERBATIM into
External-facing API calls.
    |
    v
Prompt punctuation BLEEDS INTO model output.
    |
    v
_verify_external_document_quality HARD-FAILS an External save
containing an em dash or a hyphen-pair dash.
    |
    v
*** THEREFORE Sections B and C MUST THEMSELVES contain no em dash
and no " -- ". Use commas, colons or a single spaced hyphen. ***
    |
    v
_preflight_rules checks exactly this, before a paragraph is written.

Section A is never sent, and may use them freely.

This is checked whenever B or C is edited. It is the clearest example
in the product of a rule whose FORM is constrained by its DELIVERY
MECHANISM.
```

11. Deliverable specifications

11.1 Company Charter — Internal

Attribute	Specification
Filename	<code>Company_Charter_<ProjectName>_<REG_NO>_Internal.pdf</code> (.docx retained)

Attribute	Specification
Location	<code>output/company_charters/</code> — and nowhere else
Classification	<i>Internal — Integrow Asset Management</i>
Audience	Underwriting team
Citation style	Self-descriptive inline (<code>label</code>)
Sources list	Full bibliographic, with <code>[published: ..., accessed: ...]</code> and internal filenames
Contains	Process failures · Document Library table · Weight columns in both score tables · Source Trust Registry Updates · version log (elapsed / cost / call count) · all gaps · "(Code-Computed)" heading suffixes
Section order	The ten locked sections (§11.3)
Gate	Preflight + blocking compliance review

11.2 Company Charter — External

Attribute	Specification
Filename	<code>Company_Charter_<ProjectName>_<REG_NO>_External.pdf</code>
Classification	<i>Strictly Private and Confidential</i>
Audience	Counterparty / co-investor
Citation style	Numbered <code>[N]</code> , one per distinct claim, at clause end
Sources list	Descriptive citations only, built from tokens actually cited, in first-use order
Excludes	Every process failure · file paths, module names, JSON keys, exception strings · the Document Library table · Weight columns · Source Trust Registry Updates · the version log · gaps that did not earn an Imminent or Structural flag · "(Code-Computed)" suffixes · 8 named internal-only paragraphs
Forbidden characters	Em dash, hyphen-pair dash — hard-gated
Gate	Preflight + 11-check mechanical gate + blocking compliance review

11.3 The ten locked sections

```

1. (title block)
2. Overview & Flags          <-- the reader's first page:
                               deal snapshot, headline
                               scorecard, imminent /
                               structural / monitor lists
3. Executive Summary        <-- + 4 KPI cards
4. Counterparty
5. The Asset                <-- H2: FSI (Floor Space Index)
6. Compliance & Legal Detail <-- H2: Complaint Order Outcomes,
                               Appeal-Level Judgments
7. Market & Area Intelligence
8. RERA Core Data
9. Diligence Appendix       <-- H2: Document Library Contents,
                               Credit Rating Check, IBBI
                               Insolvency Check, Company
                               Registration Profile, Group /
                               Affiliated Companies, Developer
                               Score, Documentation Authenticity
                               & Confidence Summary
10. Gaps & Sources          <-- H2: Sources

PLUS two CONDITIONAL appends that render only when their check ran:
    Land Record Check -- Maha Bhulekh Property Card
    Review Authenticity Triage (Code-Computed Heuristics)

*** SECTION ORDER IS LOCKED. Reshape content inside the flow;
    never reorder or rename a section. ***

```

11.4 .facts.json — the record

Attribute	Specification
Filename	Company_Charter_<ProjectName>_<REG_NO>.facts.json
Classification	Internal, unscrubbed. Never leaves the system.
Contents	28 required model-authored keys + ~20 code-computed keys, with clean-check text and process-failure text restored
Purpose	The complete record; the input to charter_report.py, executive_briefing.py, finalize_report.py, cts_resolve.py; the basis of the free no-API render loop
Invariant	developer_score in this file is an output, never a valid input

11.5 RERA project summary PDF

Attribute	Specification
Filename	output/<REG_NO>/<REG_NO>_summary.pdf
Renderer	ReportLab (no native dependencies)
Sections	Cover (+ data-reliability note) → Project Details → Company Charter Highlights → Promoter Profile → Market Research → remaining categories
Rebuildable	Yes, offline, zero network calls
Note	A different document from the Charter. Do not confuse them.

11.6 Supporting artefacts

Artefact	Purpose
Company_Charter_<REG>_claude_md_review.json	The compliance review's verdict, violations and unverifiable items
usage_summary.json	Per-label calls, turns, tokens and USD
output/usage_log.jsonl	Append-only, one rollup line per run, forever
documents_manifest.json	Every document with status, method, document_id, source_filename
complaint_orders_manifest.json	As above, plus complaint_id and complaint_registration_no so an outcome can be attributed back to a specific complaint
run_meta.json	reg-no, project id, auth source, promoter name, charter path, timestamp
promoter/portfolio.json	Portfolio totals, per-project rows, and the standing limitations list
research/deep_research.json	Three blocks, stamped _generated_at and _reused_prior
source_trust_registry.json	Committed to the repository — cross-run open-web trust tallies

12. Non-functional requirements

12.1 Reliability (FR-NFR-REL)

ID	Requirement
NFR-REL-01	A failure in any optional stage MUST NOT prevent the run from completing
NFR-REL-02	A failure in one category, document, or promoter-portfolio project MUST NOT affect any other
NFR-REL-03	A refusal or gate failure MUST NOT leave a half-written .docx on disk

ID	Requirement
NFR-REL-04	Corrupt cached state (token cache) MUST degrade to "absent", never to an exception
NFR-REL-05	Editorial passes MUST be idempotent — safe to run across two renders
NFR-REL-06	Re-running the same project MUST NOT silently clobber the previous run

12.2 Performance (NFR-PERF)

ID	Requirement
NFR-PERF-01	Category fetch MUST be parallelised (8 workers)
NFR-PERF-02	Document download MUST be parallelised (8 workers)
NFR-PERF-03	The five external corporate checks MUST run in parallel (5 workers)
NFR-PERF-04	Documents already downloaded MUST be reused rather than re-fetched from MahaRERA's slow DMS
NFR-PERF-05	Confirmed research sources MUST be reused within 24 hours; only open gaps re-attempted
NFR-PERF-06	Section B MUST be delivered as a cacheable system-prompt prefix block
NFR-PERF-07	All human-attended browser work MUST be contiguous in the pipeline, so the operator is not called back after a long unattended stage
NFR-PERF-08	Rendering MUST be verifiable with zero API calls from a saved <code>facts.json</code>

12.3 Usability (NFR-USE)

ID	Requirement
NFR-USE-01	Every console line MUST be prefixed <code>[INFO]</code> , <code>[OK]</code> , <code>[WARN]</code> or <code>[ERROR]</code>
NFR-USE-02	A failed stage MUST print the exact command to retry just that stage
NFR-USE-03	The CAPTCHA wait MUST print a progress line at least every 30 seconds
NFR-USE-04	The run summary MUST show cost broken down by label, sorted by cost
NFR-USE-05	Endpoint trust levels MUST be surfaced to the operator, and on the summary PDF cover
NFR-USE-06	Truncation, capping or sampling MUST NEVER be silent

12.4 Maintainability (NFR-MAINT)

ID	Requirement
NFR-MAINT-01	Guards MUST be documented by <code>module.symbol</code> , never by line number
NFR-MAINT-02	A named guard symbol disappearing MUST fail the test suite
NFR-MAINT-03	Content rules MUST live in one file that is both human-readable and machine-parsed
NFR-MAINT-04	A completed spec MUST be marked historical, with its specific staleness enumerated
NFR-MAINT-05	A module retained only as a shared library MUST say so, and say who imports what
NFR-MAINT-06	Every guard that "looks redundant" MUST record why it exists

12.5 Portability (NFR-PORT)

ID	Requirement
NFR-PORT-01	Everything except PDF conversion MUST run on Linux, macOS and Windows
NFR-PORT-02	PDF conversion currently requires Windows + Word (docx2pdf); the .docx MUST remain as the fallback output
NFR-PORT-03	Windows-only tests MUST be skippable so the suite runs on Linux/CI
NFR-PORT-04	OCR language data MUST be bundled in-repo, not assumed installed
NFR-PORT-05	The summary PDF renderer MUST avoid native dependencies, to keep a hosted-service path open

12.6 Security and privacy (NFR-SEC)

ID	Requirement
NFR-SEC-01	API keys MUST be read from the environment and MUST NOT be written to disk or logged
NFR-SEC-02	All run output MUST be gitignored
NFR-SEC-03	The External document MUST NOT contain any internal tooling detail
NFR-SEC-04	.facts.json MUST be treated as Internal-classification and never forwarded to a counterparty
NFR-SEC-05	The system MUST NOT solve or read a CAPTCHA image
NFR-SEC-06	Third-party rate limits and politeness policies MUST be honoured (Nominatim ~1 req/s with an identifying User-Agent)
NFR-SEC-07	Any integration with a Terms-of-Service question MUST be opt-in, off by default, and carry an in-code caveat

13. Governance, quality and compliance requirements

13.1 The gate chain (FR-GATE)

```

                facts assembled
                |
+-----+-----+-----+-----+
| GATE 1  PREFLIGHT                                     |
| Runs BEFORE a paragraph is written.                  |
| rules.md present . markers intact . sections non-empty |
| . B (and C when external) free of em dash and " -- " |
| . Section A validated but NOT RETURNED                |
+-----+-----+-----+-----+
| pass
| render Internal, then External
|
+-----+-----+-----+-----+
| GATE 2  EXTERNAL QUALITY  (11 mechanical checks)      |
| Re-opens the SAVED file. Blocks the SAVE.            |
| The headline check: A CITATION MARKER ATTACHED TO AN |
| ABSENCE -- the specific defect the whole editorial   |
| policy exists to prevent.                            |
+-----+-----+-----+-----+
| pass
+-----+-----+-----+-----+
| GATE 3  COMPLIANCE REVIEW  *** BLOCKS THE PDF ***    |
| Re-reads BOTH saved documents and audits them against |
| rules.md via the model.                               |
|                                                       |
| STRICT BY DEFAULT. Two ways to fail:                 |
| (a) a VERIFIED violation                             |
| (b) a review that COULD NOT RUN AT ALL                |
| -- unverified is not the same as clean               |
|                                                       |
| Safe to block on a model's judgement ONLY because every |
| violation is verified against the document text first. |
| An invented quote is logged unverifiable and CANNOT  |
| stop a good document. Quotes under 12 chars are not  |
| trusted -- they match almost anything by coincidence. |
|                                                       |
| The model's role is reduced from "decide whether this |
| complies" to "POINT AT the offending text", and the   |
| pointing is then checked mechanically.                |
+-----+-----+-----+-----+
| pass
| convert both to PDF
|
*** THE DELIVERABLE ***

```

ID	Requirement
FR-GATE-01	Preflight MUST run before any content is written and MUST fail the run on a missing, renamed or empty rules section
FR-GATE-02	Preflight MUST validate Section A without returning it, so it cannot become a route by which coding-time guidance reaches an API call
FR-GATE-03	The External mechanical gate MUST re-open the saved file and run all eleven checks, walking body paragraphs and every table cell recursively
FR-GATE-04	The compliance review MUST block the PDF by default
FR-GATE-05	A review that could not run MUST count as a failure
FR-GATE-06	Every reported violation MUST be verified against the document text before it may block; quotes under 12 normalised characters MUST NOT be trusted
FR-GATE-07	Explicitly permitted patterns ("Nothing found.", Gap N. , score-note absences, identity-table identifiers, "N/A") MUST be carved out
FR-GATE-08	Section A MUST NEVER be transmitted to any API, and a test MUST assert this — asserting the marker is present <i>before</i> asserting it is absent

ID	Requirement
FR-GATE-09	An override (CHARTER_ALLOW_UNCHECKED) MAY exist, and MUST be documented as <i>a decision to ship an unchecked document, not a convenience</i>
FR-GATE-10	The cost of the review MUST be separately labelled so it is visible and arguable

13.2 Degradation requirements (FR-DEG)

ID	Requirement
FR-DEG-01	Every model-backed judgement MUST keep its deterministic predecessor behind it
FR-DEG-02	Fallback caches MUST be keyed such that a lookup miss is indistinguishable from "no model ran"
FR-DEG-03	Malformed model replies MUST be discarded, not trusted: out-of-range ids, unrecognised verdict kinds and null matches are all dropped
FR-DEG-04	A per-finding research failure MUST cost only its own finding
FR-DEG-05	No finding may ever be deleted by any research outcome, including an empty or malformed reply
FR-DEG-06	Fan-out MUST be bounded, and the bound MUST be documented
FR-DEG-07	Per-finding research MUST NOT recompound — an already-enriched finding must be recognised by fingerprint and skipped

13.3 Reversibility requirements (FR-REV)

ID	Requirement
FR-REV-01	Any pass that removes text from the page MUST stash the original and restore it before .facts.json is written
FR-REV-02	Without this, every run would quietly hollow out the file the next run reads. This MUST NOT be allowed to happen
FR-REV-03	A relocation is the deliberate exception and MUST NOT be reversed — nothing is lost and the corrected record is the better one

14. Data requirements

14.1 The facts contract

ID	Requirement
FR-DATA-01	The facts schema MUST require, for every sourced field, both a value and a source — a sourced field cannot exist without its source
FR-DATA-02	Sources MUST carry an enumerated topic from a fixed list of 13, driving both corroboration counting and automatic citation resolution
FR-DATA-03	Code-computed fields (document library, portfolio, complaint outcomes, all check results, both scores) MUST be attached verbatim and MUST overwrite any model output for the same key
FR-DATA-04	An unconfirmable optional structure (developer_track_record) MUST be omitted entirely rather than filled with nulls
FR-DATA-05	Pre-RERA intake files MUST use the exact facts.json keys, so a later merge is a plain dict update and never a reshape
FR-DATA-06	Ephemeral render-time keys MUST be stripped before persist

14.2 Retention and lifecycle

ID	Requirement
FR-DATA-07	Prior runs MUST be archived, not overwritten
FR-DATA-08	Prior Charters MUST be archived too, matched by registration-number suffix (the project-name segment can change when a RERA record is renamed)
FR-DATA-09	Field-level change between runs (e.g. mortgage lender) MUST be detected and surfaced
FR-DATA-10	The cost ledger MUST be append-only
FR-DATA-11	The source-trust registry MUST accumulate across all projects and SHOULD be version-controlled so promotions are reviewable
FR-DATA-12	[GAP] A retention policy for <code>output/_history/</code> and PII-bearing <code>gst_portal_raw/</code> MUST be defined. <i>Not yet implemented — see §19</i>

14.3 Data quality invariants

INVARIANT	ENFORCED BY
-----	-----
A failed geocode is DROPPED, never counted as "0 km away"	<code>promoter_portfolio</code>
An unresolvable GST frequency is EXCLUDED, never given a guessed due date	<code>gst_compliance</code>
A period not yet due is EXCLUDED from every count -- not a gap, just not due yet	<code>gst_compliance</code>
A verification that COULD NOT RUN keeps the source and adds an honest gap	<code>deep_research._verify_block</code>
Two disagreeing registries BOTH surface	<code>_merge_director_rosters</code>
Two known data contradictions are DELIBERATELY left unreconciled	<code>rules.md</code> Section A
The subject project is excluded from its own portfolio totals by the ENTRY's own reg-no, never by the fetch reg-no	<code>promoter_portfolio</code>
An empty GST records file is NEVER written -- the file's existence means "data supplied"	<code>gst_intake</code>

15. Constraints, assumptions and dependencies

15.1 Hard constraints

ID	Constraint	Consequence
C1	7 of 9 MahaRERA categories are CAPTCHA-gated	A human is required, or a token supplied
C2	Maha Bhulekh admits no reusable session	Land records are opt-in and manual, four steps plus a solve
C3	The GST portal needs one solve per PAN search and one per GSTIN	GST is opt-in, and sits beside the other browser work
C4	<code>docx2pdf</code> is Windows-only	The deliverable requires Windows + Word
C5	Marathi land-record labels do not map to RERA's English text	Office/village must be human-picked; never fuzzy-matched

ID	Constraint	Consequence
C6	<code>output/</code> is gitignored and holds the <code>.docx</code> template	The template must be backed up manually before structural change
C7	Sections B and C are injected verbatim into prompts	They must themselves avoid the characters the gate forbids
C8	The External document is generated entirely by code	Editing the rules alone cannot change External behaviour; it must be implemented in code
C9	Sonnet 5 intro pricing expires 2026-08-31	Cost figures under-report by a third after that date unless updated
C10	No packaging manifest, no CI	The suite is run by hand; drift is invisible until someone runs it

15.2 Assumptions

ID	Assumption	Risk if wrong
A1	MahaRERA's public record is materially complete and current	The whole product's evidence base weakens
A2	An operator is available to solve CAPTCHAs during a run	Full runs become impossible; only 2 of 9 categories fetchable
A3	The MCA mirrors continue to publish freely	The 3-way cross-check degrades toward single-source
A4	RERA document sets always include a PAN and never a GSTIN	<code>--pan</code> remains the usual flag; <code>--gstn</code> the exception
A5	Word is installed on the run host	No PDF — the <code>.docx</code> fallback becomes the deliverable
A6	Positional parsing of MahaRERA search cards remains valid	Candidate metadata degrades (but <code>raw_text</code> is always retained)

15.3 Dependencies

Dependency	Version	Used for	Criticality
<code>playwright</code>	≥1.44	All browser work	Critical
<code>requests</code>	≥2.31	All HTTP	Critical
<code>anthropic</code>	≥0.40	Every LLM call, via <code>tool_runner</code>	Critical
<code>python-docx</code>	≥1.1	Both Charter builders + Executive Briefing	Critical
<code>docx2pdf</code>	≥0.1.8	The deliverable	Critical (Windows)
<code>reportlab</code>	≥4.0	Summary PDF	High
<code>pymupdf</code>	≥1.24	Document text extraction	High
<code>pytesseract</code> + <code>Pillow</code>	≥0.3 / ≥10.0	OCR fallback	Medium
<code>beautifulsoup4</code>	≥4.12	Registry and land-record scraping	High
<code>streamlit</code>	≥1.32	UI	Low
<code>matplotlib</code>	undeclared	Executive Briefing donut charts	Gap — see §19
Tesseract binary + <code>eng/mar</code> data	—	OCR	Medium (bundled data)
Microsoft Word	—	COM automation for PDF	Critical (Windows)

16. Out of scope

Item	Rationale
States other than Maharashtra	Different portals, schemas, land systems, languages
Continuous monitoring / alerting	Batch, on-demand, human-triggered by design
Automated CAPTCHA solving	Hard ethical and ToS boundary
Legal title opinion	The land-record check is corroborative; a Title Report is a lawyer's product
Financial modelling, valuation, IRR	Downstream of this product
Multi-tenant hosting, RBAC, audit login	Single-operator local tool
A REST/JSON API	No consumer requires one today
Automated identifier linking (CIN/CTS → reg-no)	Deliberately an explicit human action, never auto-matched
Reconciling the two known data contradictions	Deliberately preserved
Translating the Charter	English only
A database	The filesystem is the data store; <code>.facts.json</code> is the record

17. Acceptance criteria — definition of done

17.1 For a single run

- [] The correct builder was used:
company_charter.py::_fill_template via run_company_charter
(NOT charter_report.py, which is a different document for a
different request; charter_document.py builds nothing at all)
- [] A REAL monitor-flag resolution pass was performed -- flags in
Overview & Flags were actually re-checked against current facts,
not carried over stale
- [] The compliance review stage RAN, or its failure was REPORTED
rather than passed over silently
- [] Output exists only in output/company_charters/
- [] Both PDFs exist (or the conversion failure was reported)
- [] .facts.json was written AFTER restoration, so it holds the
complete unscrubbed record
- [] The run summary shows per-label API cost
- [] Any script that hit _fill_template directly called
_convert_docx_to_pdf itself

17.2 For the External document specifically

- [] Zero em dashes, zero " -- "
- [] Zero citation markers attached to an absence
- [] Zero uncited factual bullets (allowing the explicit exemption list: pipeline-computed figures, qualifier sentences, statements of statute)
- [] Zero markers glued mid-word
- [] Zero orphan sources (listed but never cited)
- [] Zero leftover Internal Document Library status text
- [] Zero italic placeholder styling (except the intentional red Standing Gap paragraph)
- [] Zero run colours outside the allowed set
- [] Balanced parentheses in every numbered Sources entry
- [] Bullet numbering intact on every Sources entry
- [] No Weight column in either score table
- [] Bullet hanging-indent fix present in the numbering part

17.3 For a code change

- [] python -m pytest -q passes, and any remaining failure is one you can explain -- record the count in the commit message so drift is visible next time
- [] If a guard was added: its symbol is in the right table in guardrails.md, test_guardrails_doc.py resolves it, and there is a test that BREAKS IT DELIBERATELY ("a guard nobody has watched fail is only a comment")
- [] If rules.md Section B or C was edited: no em dash, no " -- ", and _preflight_rules still passes
- [] If a rendering change was made: verified via the free no-API _fill_template loop against a saved facts.json
- [] If the template's structure was touched: a timestamped backup was taken first
- [] CLAUDE.md updated if the FLOW changed; rules.md updated if the CONTENT RULES changed

18. Release history and roadmap

18.1 Shipped (as of 2026-08-11)

- v0.x ACQUISITION
 - [x] Project resolution, guest-session management, 9-category scrape
 - [x] Document + complaint-order download with reuse
 - [x] Summary PDF, run archiving, endpoint verification mode
- v0.x INTELLIGENCE
 - [x] Agentic deep research with per-claim verification
 - [x] Bounded gap retry with differentiated strategies
 - [x] Promoter portfolio with 5 km micro-market influence
 - [x] 3-way MCA-mirror chain with roster-conflict surfacing
 - [x] Credit rating (ICRA + Infomerics), IBBI, group companies
 - [x] MahaRERA judgments search + appeal cross-referencing
- v0.x COMPLIANCE
 - [x] GST intake, confirmed live 2026-07-31 (76 scoreable periods)
 - [x] Statutory due dates incl. state-specific QRMP
 - [x] CTS land-record path with human-in-the-loop resolution
 - [x] Pre-RERA intake (CIN-only, CTS-only) + explicit attachment
- v1.0 THE RESHAPE (2026-08-10, all 7 tasks)
 - [x] Task 0 Test fixtures repaired
 - [x] Task 1 Flags become headlines, Gaps carries the detail
 - [x] Task 2 Clean checks deleted (prompt + scrubber + gate)
 - [x] Task 3 Per-clause External citations (shipped as topic matching, later moved to a model pass -- per-claim tokens in the schema were REJECTED as a breaking change needing regeneration)
 - [x] Task 4 Facts-schema corrections (merge orders relocated, mortgage fields merged, landowner row made a pointer)
 - [x] Task 5 Per-finding deep research as a pipeline stage
 - [x] Rules migrated from CLAUDE.md into rules.md
 - [x] charter_document.py builder retired; module kept
- v1.0 GOVERNANCE (2026-08-11)
 - [x] guardrails.md + test_guardrails_doc.py symbol enforcement
 - [x] Blocking compliance review with mechanical quote verification
 - [x] CHARTER_ALLOW_UNCHECKED documented as a deliberate decision

18.2 Roadmap

```

NEXT  --  HEALTH OF THE CODEBASE          effort  value  risk if skipped
+-----+
| R1  Add matplotlib to requirements.txt    XS      M    Executive |
|                                           Briefing |
|                                           fails on |
|                                           clean    |
|                                           install  |
| R2  Update Sonnet pricing before 08-31    XS      H    All cost  |
|      (or add a DATED pricing table)      | figures  |
|                                           wrong by |
|                                           one third|
| R3  pyproject.toml + CI running pytest    S      H    Drift     |
|      with docx2pdf skipped                | invisible|
| R4  Narrow the "Document Library" External S      H    Blocks a  |
|      gate check to the real placeholder   | planned  |
|      text                                 | feature  |
| R5  Persist the GST `as_of` date into     S      M    Re-render |
|      facts so a re-render is deterministic| drift    |
| R6  Fix the stale fetch_gstin_filing_table XS     L    Misleads  |
|      docstring                            | a reader |
+-----+

THEN  --  STRUCTURE
+-----+
| R7  Decompose company_charter.py (10,182 lines) in order of |
|      independence: registries.py -> editorial.py -> scoring.py |
|      -> docx_helpers.py. Keep _fill_template and run_company_charter|
|      together.                                           effort L, value H |
| R8  Resolve the "locked ten" vs two conditional appends      |
|      inconsistency -- fold them in, or amend the rule        |
| R9  Route Grok verification through the usage ledger          |
| R10 Retention policy for output/_history/ and gst_portal_raw/ |
+-----+

THEN  --  REACH
+-----+
| R11 LibreOffice headless as a PDF fallback -> unblocks Linux and |
|      containerised deployment                                   effort M, value H |
| R12 Wire the three unsourced Developer Score sub-metrics        |
|      (team_strength, financial_strength_debt) or document them as |
|      permanently manual                                          |
| R13 A scheduled canary run against a known-good project, alerting |
|      when a `confirmed` endpoint starts failing                 |
| R14 Batch mode: a list of registration numbers in, a pack each out |
| R15 Cross-project comparison view (the Executive Tracker, automated)|
+-----+

MAYBE --  requires a decision, not just effort
+-----+
| R16 A second state authority (each is a new portal, schema and |
|      land system -- effectively a new product)                 |
| R17 Change-monitoring: re-run on a schedule and alert on material |
|      field diffs (the archive + _diff_mortgage_lender machinery |
|      already exists; this is the productisation of it)          |
| R18 A hosted service (blocked by C4 until R11 lands)            |
+-----+

```

19. Risks and mitigations

#	Risk	L	I	Mitigation in place	Residual action
RK1	MahaRERA changes an endpoint or selector	High	High	confirmed/observed tiering; -- verify; envelope normalisation; raw_text always retained	R13 — scheduled canary
RK2	MahaRERA changes the CAPTCHA mechanism	Med	High	-- token manual escape hatch	Monitor

#	Risk	L	I	Mitigation in place	Residual action
RK3	Sonnet 5 intro pricing expires 2026-08-31	Certain	Med	—	R2 — 20 days
RK4	A registry mirror goes paywalled	Med	Med	<code>_looks_paywalled</code> ; 3-way chain; conflicts surfaced	Accept
RK5	Word/COM unavailable on the run host	Med	High	<code>.docx</code> retained as fallback output	R11
RK6	Template <code>.docx</code> lost or corrupted	Low	Critical	Section A mandates a timestamped backup	Automate the backup; template is gitignored
RK7	<code>company_charter.py</code> becomes unmaintainable	Med	High	<code>CLAUDE.md</code> + <code>guardrails.md</code> + <code>rules.md</code> navigation	R7
RK8	The suite rots silently without CI	High	High	28 test files exist and are meaningful	R3
RK9	A guard test passes vacuously	Med	High	Precedent already caught and fixed (Section-A test)	Review assertion preconditions on every guard test
RK10	PII accumulates indefinitely in <code>gst_portal_raw/</code> and <code>_history/</code>	Certain	Med	<code>output/</code> is gitignored	R10
RK11	GST scoring drifts on re-render (<code>as_of</code> defaults to today)	Med	Med	<code>run_gst_compliance_check</code> passes an explicit date at run time	R5
RK12	Anthropic API outage mid-run	Low	Med	Never fatal; standalone retry; <code>verification_error</code> keeps sources	Accept
RK13	An operator ships an External document with <code>CHARTER_ALLOW_UNCHECKED=1</code>	Low	High	Documented as <i>a decision, not a convenience</i> ; the review JSON records it	Consider making the override stamp the document
RK14	<code>PROMOTER_PROJECT_LIMIT = 25</code> silently understates a large developer	Med	Med	Never silent — <code>truncated</code> field plus an extra limitation	Consider raising, or paginating
RK15	Google Maps scrape is a ToS exposure	Low	Med	Off by default; in-code caveat; always falls back	Keep off; consider the paid Routes API
RK16	The two conditional sections violate the locked-order rule	Certain	Low	Documented here and in <code>SAD.md</code> §25	R8

20. Open questions

#	Question	Owner	Blocking
Q1	Should the three unsourced Developer Score sub-metrics be wired to a data source, or formally declared manual-input? Today they silently drag the composite down	Product	R12
Q2	What is the retention period for PII-bearing <code>gst_portal_raw/</code> screenshots?	Compliance	R10
Q3	Should <code>CHARTER_ALLOW_UNCHECKED</code> stamp a visible marker into the document it ships?	Product	RK13
Q4	Is LibreOffice-rendered PDF fidelity acceptable for a client-facing External document, or is Word required?	Product + Design	R11
Q5	Should the two conditional sections be folded into Diligence Appendix, or should the locked-ten rule be amended to "ten unconditional plus two conditional"?	Engineering	R8

#	Question	Owner	Blocking
Q 6	Should <code>charter_report.py</code> 's "Counterparty + Collateral" layout eventually replace the ten-section Charter, or remain a parallel deliverable? <code>rules.md</code> says its layout is locked and to <i>ask first</i>	Product	R7 scope
Q 7	Is a 24-hour research reuse window right? Too long for a fast-moving litigation picture, too short for stable macro-market data	Product	—
Q 8	Should <code>executive_briefing.py</code> get a CLI, or be formally declared library-only?	Engineering	D13

21. Appendices

Appendix A — Requirement traceability

Product goal	Functional requirements	Enforcement	Verified by
PG1 Compress diligence time	FR-ACQ-08, FR-CORP-09, FR-ACQ-15, FR-RES-08	Parallelism + reuse	Wall-clock per run
PG2 Every claim traceable	FR-DATA-01/02, FR-EXT-01..05	Schema + citation registry	Gate 2 checks 7–8; <code>test_external_citations.py</code>
PG3 Never guess	FR-RES-07, FR-TRK-06, FR-COMP-10, FR-LAND-03	Drop-not-guess everywhere	<code>test_data_quality_fixes.py</code>
PG4 Clear vs unknown	FR-ED-01/02/06	Scrubber + gate	<code>test_clean_check_scrubber.py</code>
PG5 Partial failure never costs the run	FR-CORP-10, FR-TRK-10, FR-COMP-12, FR-RES-12, NFR-REL-01/02	Never-fatal wrappers	<code>test_gst_intake.py</code> , <code>test_promoter_portfolio.py</code>
PG6 Non-compliant does not ship	FR-GATE-01..10	Three-tier gate chain	<code>test_rules_preflight.py</code> , <code>test_claude_md_doc_review.py</code>
PG7 Two audiences, one analysis	FR-DOC-01/02, FR-EXT-*	Render-time fork	<code>test_charter_report_source_labels.py</code>
PG8 Costs visible	FR-OPS-02/03/08	Per-label ledger	<code>test_usage_tracking.py</code>
PG9 Cheap re-runs	FR-ACQ-15/17/18, FR-RES-08	Archive + reuse	<code>test_attach_rera_number.py</code>
PG10 Record outlives page	FR-REV-01..03, FR-DOC-08	Scrub/restore	<code>test_process_text_sanitizer.py</code>

Appendix B — Command reference

PRIMARY

```
python main.py <REG_NO | project name>
  [--gstn X | --pan Y]    opt-in GST intake (mutually exclusive)
  [--headed]              show the search browser
  [--token T]             supply a session you captured yourself
  [--no-auto-auth]        skip the CAPTCHA browser (2 categories only)
  [--captcha-timeout N]   default 300
  [--project-id N]        bypass search resolution entirely
  [--output-dir D]        default "output"
  [--verify]              probe endpoints instead of building
```

```
streamlit run app.py          same pipeline, 3 tabs
```

COMPONENT RETRY

```
python deep_research.py <REG_NO> [--output-dir] [--no-rebuild] [--force-refresh]
python company_charter.py <REG_NO>
python finalize_report.py <REG_NO>          zero network calls
```

PRE-RERA INTAKE

```
python promoter_intake.py <CIN> [company_name]
python cts_intake.py <district> <office> <village> <cts> <mobile>
python attach_rera.py <case_id> <reg_no>
```

LAND RECORDS (in order)

```
python mahabhumi.py offices <district>
python mahabhumi.py villages <district> <office_label>
python cts_resolve.py offices <REG>
python cts_resolve.py villages <REG> "<office>"
python cts_resolve.py candidates <REG> "<office>" "<village>" <cts_query>
python cts_resolve.py finalize <REG> "<office>" "<village>" <cts> <mobile>
```

GST

```
python gst_intake.py <PAN|GSTIN> <reg_no>
python gst_portal.py pan <PAN>
python gst_portal.py filing <GSTIN>
```

SESSION

```
python token_cache.py get          exit 0 ok, 1 stale/absent
python token_cache.py set <token>
python token_cache.py minutes-left
```

SECOND BUILDER

```
python run_charter_pipeline.py      prepare/finish seam
python build_report.py              one-off, per engagement
```

Appendix C — Environment variables

Variable	Required	Effect
ANTHROPIC_API_KEY	For research + Charter	Absent → both degrade, never fatally
XAI_API_KEY	Optional	Enables the Grok second-opinion document-grounding path
TESSERACT_CMD	Optional	OCR binary location; falls back to PATH then two Windows paths
COMPANY_CHARTER_USE_MAPS_SCRAPES	Optional	Must be exactly "1". Opt-in Maps distance refinement; ToS caveat
CHARTER_ALLOW_UNCHECKED	Optional	1/t rue/yes downgrades the blocking review to advisory. A decision to ship an unchecked document, not a convenience

Appendix D — Where to look when changing something

CHANGING...	READ FIRST	THEN CHECK
-----	-----	-----
the pipeline flow what the document may say	CLAUDE.md rules.md	update CLAUDE.md _preflight_rules, then the gates add the symbol, write the test that breaks it *** ASK FIRST. The order is locked. ***
a gate, fallback or bound	guardrails.md	
the ten sections	rules.md Section A	
the .docx template structure	rules.md Section A	BACK IT UP first (timestamped)
Section B or C text	rules.md Section A	no em dash, no " -- "
External behaviour	CHARTER_RESHAPE_SPEC.md (historical note)	It is generated BY CODE. Editing rules alone will NOT move it.
scoring	SAD.md section 17	Developer Score never renormalises; Doc Confidence always does
anything in company_charter.py	SAD.md section 9	10,182 lines -- use the function group map

Appendix E — Glossary

Term	Meaning
Charter	The Company Charter document pair. The primary deliverable
Clean check	A check that ran and came back clear. Produces no sentence
Gap	An open unknown the reader can act on. Kept in full, numbered
Finding	Something established. Earns a dedicated research pass
Flag	A one-sentence headline ending in (Gap N) . Imminent / structural / monitor
Variant	internal or external — the fork at render time
Carryover	A pre-RERA intake result attached to a reg-no by explicit human action
`confirmed` / `observed`	Endpoint trust levels — individually re-verified vs taken from real traffic
Never fatal	A stage whose failure logs a warning and lets the run continue
Hard gate	A guard that stops the run or blocks a save or the deliverable
QRMP	Quarterly Return Monthly Payment — the GST scheme with state-specific due dates
CTS	City Survey number — the Maharashtra land parcel identifier
Property Card	मालमत्ता पत्रक — the Maha Bhulekh land record
SPV	Single-purpose vehicle — a company formed for one project. Detecting one is a core product job
Roster conflict	Two registries naming different directors for the same company. Both are surfaced

End of Product Requirements Document.